

Kieara Blackwood

Professor Sawant

CS 472: Introduction to Deep Learning

23 April 2026

Project 2: Deep Learning for Image Classification

Abstract

In this project, I built a deep learning model to classify images into different categories from an unstructured image dataset. This classification model was trained on labeled image data to determine which image corresponds to the specific class. Throughout this project, I preprocessed unstructured data, built a Convolutional Neural Network, trained classification models and evaluated their performance. Ultimately, the goal of this project was to design a neural network that can accurately predict the class of an image.

Introduction

Butterflies and moths are two closely related insects that share a common ancestor and have similar life cycles. On a global scale, there are about 17,500 species of butterflies and 160,000 species of moths. The goal of this project is to create a deep learning model that can classify images into predefined categories. These predefined categories will be based from 100 distinct species of butterflies and moths. Based on these categories, a Convolutional Neural Network will be designed and evaluated using Accuracy Score, Confusion Matrix, and Loss Function. Additionally, a train-test split will be implemented in order for the model to identify patterns within the dataset.

Methods

For this project I gathered my data from the online website, Kaggle. This dataset, called Butterfly & Moths Image Classification 100 species, was created by Gerry. This dataset contains labeled images of butterfly and moth species that are organized into training, validation, and test splits via separate CSV files. Additionally, this dataset covers over 100 distinct species of butterflies and moths. For the purpose of this project, I will be strictly using the butterflies and moths.csv file to train and test my model. This dataset consists of 4 columns and 12,595 rows with 12, 594 trained, 500 tested, and 500 valid data.

Model and Approach

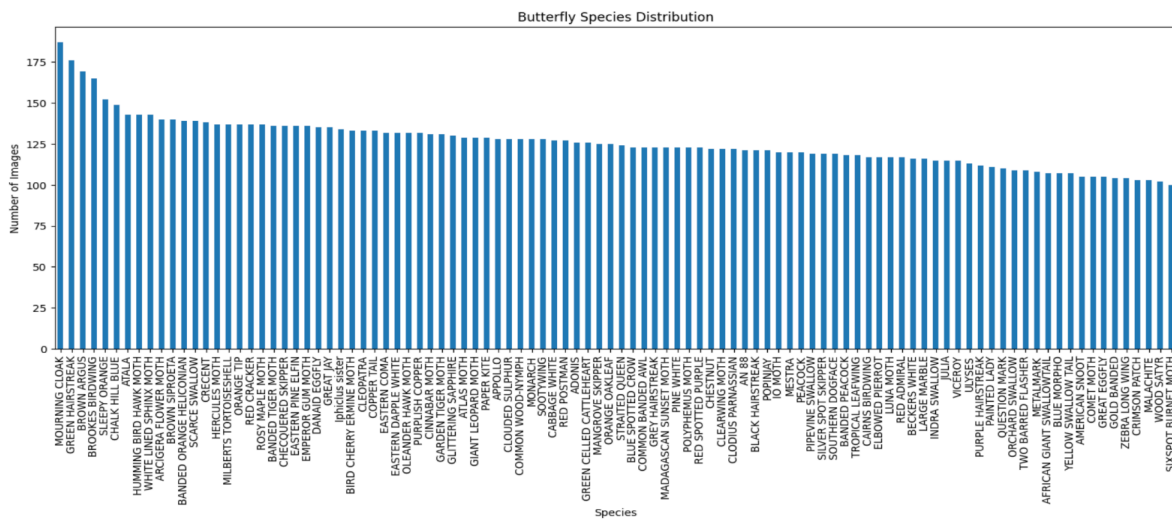
A Convolutional Neural Network was designed and trained using TensorFlow and Keras. This image classifier model contains the following layers: three convolution layers (Conv2D and MaxPooling2D) to extract spatial features, a fully connected layer and dropout layer to prevent overfitting, and a softmax output layer to produce a probability score between 0 and 1 for each class. In addition, these images were preprocessed by resizing the images (64x64 pixels) and normalizing the pixel values from [0, 255] to [0, 1]. By using normalization, it stabilizes gradient updates during training and speeds up convergence. Moreover, I also included improvement techniques such as data augmentation and dropout to my model building process. With data augmentation, I was able to diversify the training data, while with dropout (rate = 0.4), it allowed for more robust feature learning.

Results

Before Normalization



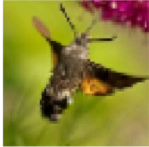
After Normalization



Total training images : 12594
 Most common class : MOURNING CLOAK (187 images)
 Least common class : SIXSPOT BURNET MOTH (100 images)

Sample Training Images

HUMMING BIRD HAWK MOTH



ATALA



BROOKES BIRDWING



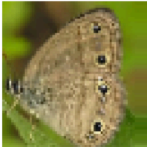
BANDED ORANGE HELICONIAN



SLEEPY ORANGE



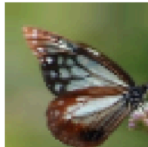
WOOD SATYR



COMMON BANDED AIL



CHESTNUT



IO MOTH



Model: "sequential"

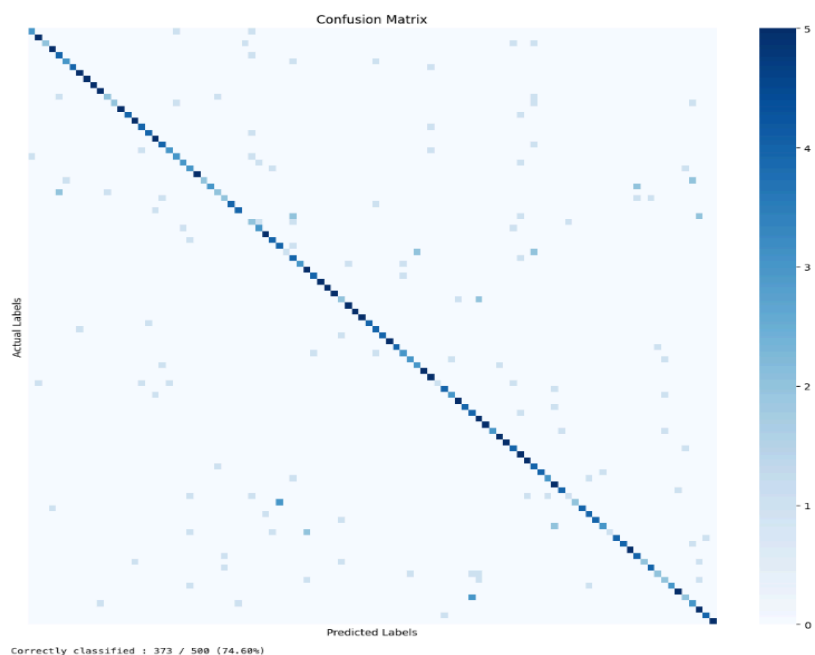
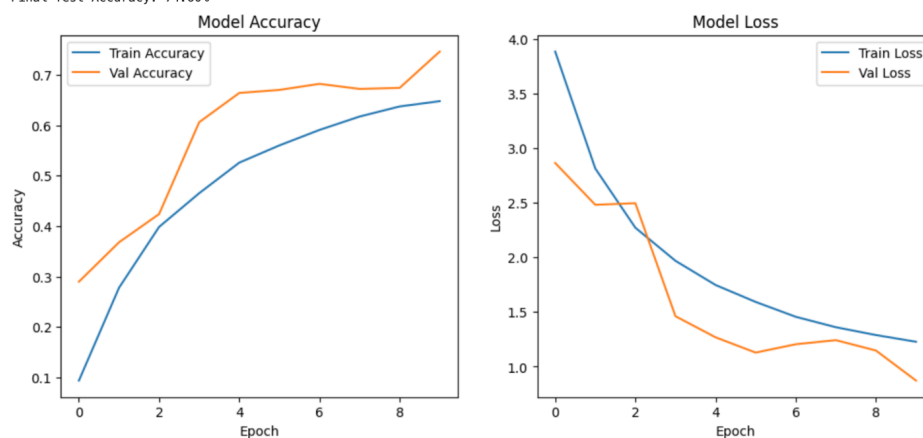
Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 62, 62, 32)	896
max_pooling2d (MaxPooling2D)	(None, 31, 31, 32)	0
conv2d_1 (Conv2D)	(None, 29, 29, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 14, 14, 64)	0
conv2d_2 (Conv2D)	(None, 12, 12, 128)	73,856
max_pooling2d_2 (MaxPooling2D)	(None, 6, 6, 128)	0
flatten (Flatten)	(None, 4608)	0
dense (Dense)	(None, 256)	1,179,904
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 100)	25,700

Total params: 1,298,852 (4.95 MB)

Trainable params: 1,298,852 (4.95 MB)

Non-trainable params: 0 (0.00 B)

16/16 0s 17ms/step - accuracy: 0.7460 - loss: 0.8731
 Final Test Loss: 0.8731
 Final Test Accuracy: 74.60%



Classification Report

	precision	recall	f1-score	support
ADONIS	0.75	0.60	0.67	5
AFRICAN GIANT SWALLOWTAIL	0.83	1.00	0.91	5
AMERICAN SNOOT	1.00	0.40	0.57	5
AN 88	0.83	1.00	0.91	5
APPOLLO	0.57	0.80	0.67	5
ARCIGERA FLOWER MOTH	0.75	0.60	0.67	5
ATALA	1.00	0.80	0.89	5
ATLAS MOTH	0.83	1.00	0.91	5
BANDED ORANGE HELICONIAN	1.00	1.00	1.00	5
BANDED PEACOCK	1.00	1.00	1.00	5
BANDED TIGER MOTH	0.83	1.00	0.91	5
BECKERS WHITE	0.67	0.40	0.50	5
BIRD CHERRY ERMINE MOTH	1.00	0.40	0.57	5
BLACK HAIRSTREAK	1.00	1.00	1.00	5

Based on the results, the model achieved a strong training and validation accuracy score over 10 rounds. After each round, the model got better at recognizing the different types of butterflies and moths. For the accuracy curve diagram, the lines for training and testing stayed closed together. This suggests that the model not only memorized the classes, but it also learned how to recognize the features. The results for the Confusion Matrix chart (373 / 500 - 74.60%) suggests that the model did a decent job of correctly identifying the species. For the times that the model failed, a possible reason for this is that the model probably mixed up two species that look almost identical to one another. Overall, the species with the brighter colors were easier for the model to identify compared to the species with darker colors. On the other hand, some strengths of the model was that CNN was the most effective way of identifying complex visual patterns in images and data augmentation and dropout allowed for more robust feature learning. Some limitations that I had was that by shrinking the image to (64 x 64 pixels) it made the images very blurry and there was an uneven amount of data for some classes. Due to the lack of images for some species, the model struggled to classify the images compared to the ones with more images for that specific species. The only thing that I would improve from the dataset is to fine-tune the dataset so that the results have a substantially higher accuracy score and less training time.

Conclusion

The CNN model learned to distinguish over 100 butterfly and moth species with strong overall accuracy. Thus, the results confirm that the model is highly effective in identifying complex visual patterns between the butterfly and moth species.

Works Cited

- Gerry. "Butterfly & Moths Image Classification 100 species." *Kaggle*, 2023,
<https://www.kaggle.com/datasets/gpiosenska/butterfly-images40-species/data>. Accessed
23 April 2026.
- Reckner, Melissa, et al. "Butterflies vs Moths: What are the differences?" *Brandywine
Conservancy and Museum of Art*, 14 March 2023,
<https://www.brandywine.org/conservancy/blog/butterflies-vs-moths-what-are-differences>.
Accessed 23 April 2026.